

Task 9: Network Vulnerability Scanning (Nmap):

1) Find Your Local IP:

```
(hari@vbox)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe94:4af1 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe94:4af1 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:fe7e:2259:3566:eba prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:94:4a:f1 txqueuelen 1000 (Ethernet)
    RX packets 7983 bytes 10023380 (9.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3802 bytes 342591 (334.5 KiB)
    TX errors 0 dropped 21 overruns 0 carrier 0 collisions 0
```

This is your target IP (example).

2) Basic Network Scan:

```
(hari@vbox)-[~]
$ nmap 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:45 EST
Nmap scan report for 10.0.2.15
Host is up (0.0000050s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds
```

A basic Nmap scan was performed to identify active hosts and open ports.

3) Scan for Open Ports:

```
(hari@vbox)-[~]
$ nmap -p- 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:47 EST
Nmap scan report for 10.0.2.15
Host is up (0.0000040s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds
```

This scans all 65535 ports.

4) Service & Version Detection:

```
(hari@vbox)-[~]
$ nmap -sV 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:49 EST
Nmap scan report for 10.0.2.15
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.61 seconds
```

Service version detection helps identify outdated or vulnerable services.

5) Operating System Detection:

```
(hari@vbox)-[~]
$ sudo nmap -O 10.0.2.15
[sudo] password for hari:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:51 EST
Nmap scan report for 10.0.2.15
Host is up (0.00038s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops
```

OS detection provides insight into the underlying operating system.

6) Aggressive Scan:

```
(hari@vbox)-[~]
$ sudo nmap -A 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:54 EST
Nmap scan report for 10.0.2.15
Host is up (0.00035s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_ http-server-header: Apache/2.4.66 (Debian)
|_ http-title: Apache2 Debian Default Page: It works
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops
```

7) Vulnerability Scanning:

```
(hari@vbox)-[~]
$ nmap --script vuln 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 11:58 EST
Nmap scan report for 10.0.2.15
Host is up (0.0000090s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-enum:
|   /test.php: Test page
|_  /server-status/: Potentially interesting folder
Nmap done: 1 IP address (1 host up) scanned in 31.80 seconds
```

8) Save Scan Results:

```
(hari@vbox)-[~]
$ nmap -A 10.0.2.15 -oN scan_result.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 12:00 EST
Nmap scan report for 10.0.2.15
Host is up (0.00011s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_ http-title: Apache2 Debian Default Page: It works
|_ http-server-header: Apache/2.4.66 (Debian)
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.12 seconds
```

```
(hari@vbox)-[~]
$ cat scan_result.txt
# Nmap 7.95 scan initiated Thu Jan 29 12:00:42 2026 as: /usr/lib/nmap/nmap --privileged -A -oN scan_result.txt 10.0.2.15
Nmap scan report for 10.0.2.15
Host is up (0.00011s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.66 ((Debian))
|_ http-title: Apache2 Debian Default Page: It works
|_ http-server-header: Apache/2.4.66 (Debian)
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Thu Jan 29 12:00:50 2026 -- 1 IP address (1 host up) scanned in 8.12 seconds
```

9) Analyze Risks:

- Open SSH → Risk of brute force
- Open HTTP → Web attack surface

- • Outdated services → Known CVEs

10) Document Findings:

Port	Service	Version	Risk
22	SSH	OpenSSH	Brute force
80	HTTP	Apache	Web attacks